

Mustafa Burak BAŞAR

 Google Drive Portfolio

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 Konya, Turkey

 Transcript



Education

Oct 2022 – Expected 2027

■ **İzmir Democracy University, İzmir, Turkey**

Bachelor of Science in Electrical and Electronics Engineering (100% English)

GPA: 3.02 / 4.00

Focus Areas: Deep Learning (CNN), Data Mining, Computer Vision, Image Processing, DSP, Renewable Energy, Power Systems, SCADA, Machine Learning, Signal and Embedded Systems.

Skills

Languages	■ Turkish (Native), English (B2 – Professional Working Proficiency)
Programming	■ Python, C/C++, Embedded C, SQL
Machine Learning & AI	■ PyTorch, TensorFlow, Scikit-Learn, Computer Vision, Deep Learning, Image Processing, Data Mining, Data Analysis
Embedded Systems	■ STM32, ESP32, Raspberry Pi, Pixhawk, Arduino, Microcontrollers, Sensor Integration
Communication Protocols	■ UART, SPI, I2C
Operating Systems & Version Control	■ Linux (Ubuntu), Git, GitHub
Software & Development Tools	■ MATLAB, Jupyter Notebook, PyCharm, Anaconda, Proteus, Wokwi, Tinkercad, SolidWorks, AutoCAD, SketchUp, Microsoft Office
Databases	■ MySQL, SQL
Domains	■ Autonomous Systems, Embedded Systems, Computer Vision, Machine Learning, Signal Processing, Image Processing, System Integration, SCADA, Renewable Energy, Power Systems

Experience

Internships

Jul. 2025 – Sep. 2025

■ **Kalyon Energy**, Karapınar, Turkey – *Intern (PV Solar Power Plant)*

Gained hands-on experience in SCADA systems, field operations, and substation (trafo center) monitoring. Observed and assisted in troubleshooting processes, inverter systems, and data analysis for power generation efficiency.

Jul. 2026 – Aug. 2026

■ **Okumuş Elektrik Engineering**, Konya, Turkey – *Electrical Engineering Intern*

Participated in electrical project design and technical drawing activities. Gained practical experience in photovoltaic (PV) power plant projects, low-voltage (LV) and medium-voltage (MV) electrical installations, and electrical infrastructure systems.

Experience (continued)

Engineering Experience

- 2025 – Present  **İnciraltı Teknofest Team** – *Autonomous Navigation and Vision System Developer*
Developed computer vision and autonomous navigation algorithms for an autonomous unmanned ground vehicle. Implemented YOLOv8-based traffic sign detection and integrated Jetson Nano and Pixhawk platforms for real-time embedded AI applications.

Club Experiences

- 2023 – Present  **Project Development Club**, İzmir Democracy University – *Active Member*
Contributed to the design, prototyping, and presentation stages of automation and control systems projects.
- Oct. 2023 – 2024  **CIGRE Student Club**, İzmir Democracy University – *Member*
Participated in collaborative projects and workshops focused on power systems, smart grids, and SCADA applications.

Projects

Team Projects

- Teknofest 2026  **Autonomous Unmanned Ground Vehicle (UGV) – “MiHenk”**
Role: Autonomous Navigation and Vision System Developer (İnciraltı Teknofest Team)
Developed an autonomous navigation system integrating Pixhawk and Jetson Nano Edge AI platforms. Implemented YOLOv8-based traffic sign detection and optimized MAVLink communication for real-time autonomous driving applications.
Technologies: Python, YOLOv8, OpenCV, Jetson Nano, Pixhawk, MAVLink, Roboflow.
- İnciraltı Team  **UAV and Drone Systems Development**
Role: Hardware Integration and Flight Mechanics
Participated in flight mechanics analysis, subsystem design, and hardware integration processes of unmanned aerial vehicles.
Technologies: Pixhawk, Embedded Systems, UAV Systems, Sensor Integration.

Academic and Individual Projects

- Composite Design  **Glass-Fiber Reinforced Wind Turbine Design**
Advisor: Suphi Yertum (3D Mold Design Instructor)
Participated in the 3D modeling, mold design, and manufacturing processes of a glass-fiber reinforced composite wind turbine.
Technologies: SolidWorks, Composite Manufacturing, CAD Design.
- Deep Learning  **Hybrid AI-Supported Hand Gesture Control System**
Developed a real-time gesture recognition system combining MediaPipe hand landmark detection and a custom CNN architecture. Achieved over 95% recognition accuracy with approximately 180–200 ms latency, enabling wireless mouse control and application launching.
Technologies: Python, TensorFlow, OpenCV, MediaPipe, PyAutoGUI, Tkinter.
-  **Multi-Label Lung Disease Diagnosis**
Developed a DenseNet121-based deep learning model for automated multi-label chest X-ray diagnosis using the NIH ChestX-ray14 dataset and achieved class-wise AUC scores up to 0.924.
Technologies: Python, PyTorch, DenseNet121, Pandas, Albumentations.

Projects (continued)

Computer Vision	 Image Enhancement and Restoration with U-Net Developed a U-Net-based model for artifact removal, contrast restoration, and image denoising. Achieved approximately 0.89 SSIM and 18 dB PSNR after training. Technologies: Python, PyTorch, U-Net, OpenCV.
Automation	 STM32-Based Robotic Arm with Memory Algorithm Designed a robotic arm with manual and autonomous control capabilities and implemented a memory-based algorithm enabling the system to record and replay manual movements. Technologies: STM32, Embedded C, Servo Motors, PWM.
IoT and C++	 IoT-Based Smart Parking Barrier System Developed an IoT-enabled smart parking system utilizing ESP32, ultrasonic sensors, and infrared sensors with a web-based interface for manual and automated barrier control. Technologies: ESP32, C++, IoT, Web Server, Ultrasonic Sensors.
Embedded C	 STM32-Based Smart Irrigation System Designed an automated irrigation system optimizing water consumption by processing soil moisture and environmental data through an STM32 microcontroller with automatic and manual operating modes. Technologies: STM32, Embedded C, ADC, I2C, Sensors, Automation Systems.
Data Mining	 Istanbul Public Transportation Optimization Analyzed Istanbul municipality open datasets to identify peak usage periods and improve transportation efficiency using K-Means clustering algorithms. Determined the optimal number of clusters ($k = 3$) and achieved a silhouette score of 0.609. Technologies: Python, Pandas, Scikit-Learn, K-Means.
Portfolio	 Project documentations, reports, certificates, and source codes are available via the Google Drive portfolio link in the header.

Certifications

Defense and Aerospace

-  **UAV Pilot Certificate (IHA-1) – DGCA**
-  **Aviation Systems Training – Ege University**
-  **Fundamentals of Rocketry – Defense Industry Academy**
-  **TEI Motor School – Turkish Engine Industries (TEI)**

Artificial Intelligence and Software

-  **Artificial Intelligence Training – İş Bankası**
-  **Machine Learning with Python – BTK Academy**
-  **STM32 Embedded Software Development – BTK Academy**
-  **Digital Signal Processing (DSP) – Erciyes University**
-  **Unity Game Development Training – ODTÜ Bilgeİş**

Energy and Engineering

-  **Photovoltaic (PV) Power Systems – IEEE Ege University Student Branch**
-  **Composite Product Manufacturing – Republic of Türkiye Ministry of National Education**
-  **Sheet Metal Part and Mold Design – Republic of Türkiye Ministry of National Education**
-  **Computer-Aided 3D Design – Republic of Türkiye Ministry of National Education**